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Original article

Evaluation of blood donors questionnaire in a developing country: The case of Gabon

Évaluation du questionnaire de présélection de donneurs de sang dans un pays en développement : le cas du Gabon

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Abstract

Background. – Blood transfusion is a life-saving therapeutic act without alternative. Each blood transfusion carries the risk of blood-borne pathogens transmission. The present study, is the first establishing the usefulness of blood donors screening questionnaire in the setting of the Gabonese National Blood Transfusion Center.
Study design. – Nine hundred and thirty-four blood donors aged between 18 and 48 years old were initially enrolled and submitted to physical examination (body-mass index and blood pressure). After physical examination 854 donors were judged fit for blood donation and were randomly distributed in two groups. The first group of donors did not take the screening questionnaire; whereas the second group went through the screening questionnaire. Both groups were then tested for human immunodeficiency virus, hepatitis B, hepatitis C and syphilis.
Results. – Data revealed a seroprevalence among the donors of 2.5%, 2.5%, 1.1%, and 3.3% for the human immunodeficiency virus, hepatitis B, hepatitis C and syphilis markers respectively. In the Gabonese setting, blood donors' screening questionnaire reduced respectively by 0.6%, 0.35% and 1.3% the proportion of hepatitis B, hepatitis C and treponema pallidum seroreactive donors being selected for donation. The questionnaire had no positive effect on discriminating human immunodeficiency virus positive donors.
Conclusion. – Blood donors' seroprevalence of blood-borne pathogens is relatively important in our setting. Blood donors' screening questionnaire reduced the proportion of hepatitis B, hepatitis C and syphilis seropositive blood donors. The questionnaire did not effectively discriminate human immunodeficiency virus-infected donor candidates.
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Keywords: Transfusion-transmissible infections; Blood donors; Questionnaire; Developing country

Résumé

Contexte. – La transfusion sanguine est un acte thérapeutique sans alternative. Chaque transfusion sanguine comporte un risque de transmission de pathogènes à diffusion hématogène. La présente étude est la première traitant (1) de la séroprévalence des marqueurs du virus de l'immunodéficience humaine (VIH), l'hépatite B, l'hépatite C et la syphilis des donneurs de sang au gabon ; (2) de l'utilité du questionnaire de sélection des donneurs de sang dans le contexte Gabonais.
Méthode. – Neuf cent trente-quatre donneurs de sang âgés entre 18 et 48 ans ont été initialement recrutés et soumis à un examen physique (indice de masse corporelle et pression artérielle). Après examen physique, 854 donneurs jugés propres au don de sang ont été répartis aléatoirement en deux groupes. Le premier groupe de donneurs n'a pas été soumis au questionnaire de sélection, tandis que le second groupe est passé par le questionnaire de sélection. Les deux groupes ont ensuite été testés pour le VIH, l'hépatite B, l'hépatite C et la syphilis, mais pas pour le VIH.

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Résultats. – Les données ont révélé une séroprévalence chez les donneurs de 2,5 %, 2,5 %, 1,1 % et 3,3 % pour le VIH, l'hépatite B, l'hépatite C et la syphilis marqueurs respectivement. Dans le contexte Gabonais, questionnaire de sélection des donneurs de sang réduits respectivement de 0,6 %, 0,35 % et 1,3 % la proportion de donneurs séronégatifs pour l'hépatite B, l'hépatite C et treponema pallidum. Le questionnaire n'a eu aucun effet positif sur la discrimination des donneurs VIH positifs.
Conclusion. – la séroprévalence de pathogènes à diffusion hématogène chez les donneur Gabonais est relativement importante. Le questionnaire de sélection des donneurs de sang réduit la proportion des donneurs séropositifs pour l'hépatite B, l'hépatite C et la syphilis, mais pas pour le VIH.
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Mots clés : Infections transmissibles par la transfusion ; Donneurs de sang ; Questionnaire ; Pays en développement

1. Introduction

Blood transfusion remains an essential therapeutic act without alternative. With each transfusion there is a risk of blood-borne pathogens transmission, including human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and syphilis [1,2].

In developing countries blood transfusion is still challenging considering the high prevalence of blood-borne pathogens and the difficulties in ensuring safe blood supply [2–4]. The implementation of reference screening tests like nucleic acid testing (NAT) or enzyme immunoassays (EIA) for all collected blood is still an exercise beyond reach for these countries with limited resources. Donor selection is crucial in improving safe blood collection. Selection of blood donors with a low risk for transfusion-transmissible infections is the essential part of the process [5–7]. Donor questionnaire has been implemented in some countries for years and is thought to improve blood recipient safety [6–9].

In sub-Saharan Africa, although the medical history screening questionnaires are also used [10,11], countries evaluations of its effectiveness in order to develop a more cost-effective donor selection process are scarce.

The present study aimed to establish the usefulness of blood donors screening questionnaire in the setting of the Gabonese National Blood Transfusion Center.

2. Material and methods

The present study was carried out at the Gabonese National Blood Transfusion Center (CNTS) in Libreville where routine donor selection process includes:

- a physical examination;
- a screening questionnaire;
- blood tests for blood-borne pathogens.

Study design is described in Fig. 1. Briefly, 934 first time blood donors aged between 18 and 48 years old were initially enrolled for this study. Seventy-six were excluded after the physical examination (body-mass index and blood pressure). The remaining 854 were randomly distributed in two groups. To the first group of donors, no questionnaire was administered whereas the second group we administered a questionnaire. Questionnaire defferal criteria are listed in Table 1. To ensure that respondents produce honest responses, it was explained to them that the CNTS collects personal health information and safeguards its confidentiality and privacy in accordance with health care workers patients' confidentiality principles. Both groups were then tested for human immunodeficiency virus (HIV), hepatitis B (HBV), hepatitis C (HCV) and Siphilis (TPHA). Donors HIV status was assessed using Genscreen Ultra HIV Ag-Ab from Bio-Rad (Marnes-la-Coquette, France). HBV and HCV diagnosis was done using Monolisa HBS ag Ultra and Monolisa HCV ag Ultra from Bio-Rad (Marnes-la-Coquette, France). TPHA diagnosis was done using Siphilis EIA II total Antibody (Bio-Rad, Marnes-la-Coquette, France).

2.1. Ethical issue

All participants were duly informed on the study and gave their consent before entering the study. For the study The National Blood Transfusion Centre procedures were implemented in term of counseling and standard care.

3. Results

Of the 444 donors who took blood donors screening questionnaire 83 were excluded by it and 361 selected for blood donation. Thus the exclusion rate was 18.7%, which was higher than the 8.1% physical examination exclusion rate.

3.1. Seroprevalence of transfusion-transmissible transfusion-transmitted infectious diseases (TTIs)

Comparing the donors selected after questionnaire and the donors not questioned for HIV, HBV HCV and TPHA seroreactivity, data showed among donors who undertook the questionnaire a seroprevalence of 2.5%, 2.5%, 1.1%, and 3.3% for HIV, HBV, HCV and TPHA markers, respectively. Donors who were not asked to take the questionnaire showed a prevalence of 1.45%, 3.1%, 1.45% and 4.6% for HIV, HBV, HCV and TPHA markers, respectively. Undetermined results were 1.44% higher in donors who were not submitted to the questionnaire compared in donors selected after questionnaire. The rates of undetermined test results were: 1.94% in donors screened by the questionnaire and 3.38% in donors not submitted to the questionnaire.

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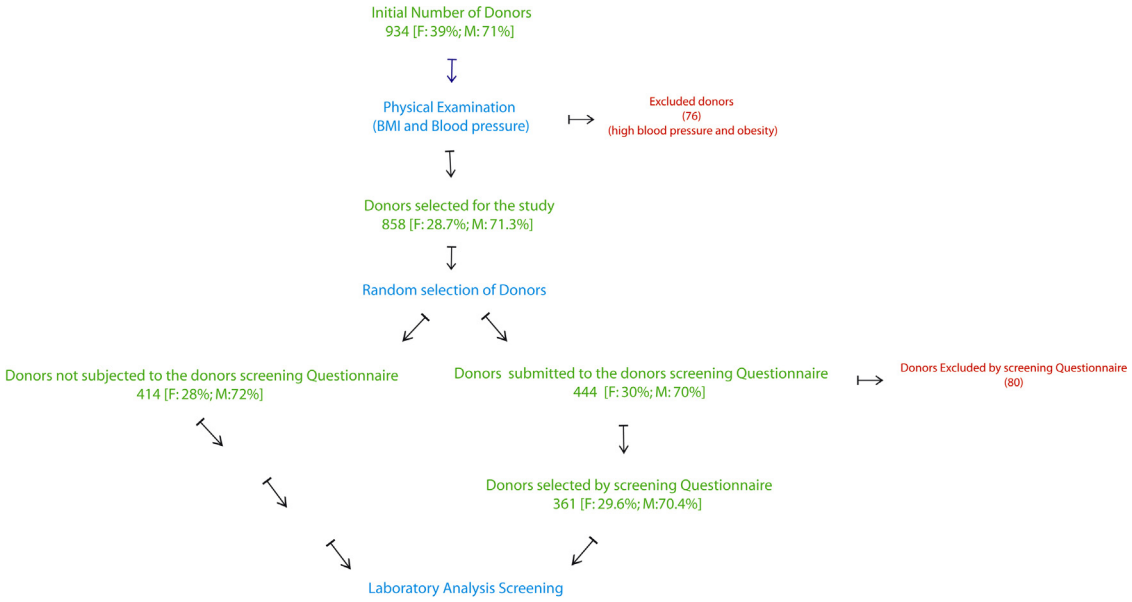


Fig. 1. Study design, 934 donors 71% male and 39% female were initially enrolled and submitted to physical examination. After physical examination the 854 donors judged fit for blood donation were randomly distributed in two groups. The first group of 414 donors did not take the screening questionnaire; the second group of 444 donors took the screening questionnaire. The questionnaire excluded 83 donors. Both groups were tested for human immunodeficiency virus (HIV), hepatitis B (HBV), hepatitis C (HCV) and syphilis.

Table 1
Resumed Gabonese screening questionnaire for blood donation and reasons for deferral in this study.

Questionnaire	Total number of persons excluded: 80
Transfusion and surgical history	8
1. Were you refused before to donate blood?	0
2. Are waiting to be operated?	0
3. Recently donated blood?	7
4. Were you recently involve in an accident?	0
5. Were you operated in the last 6 months?	0
6. Were you recently blood transfused?	1
Infectious and transmissible diseases	4
7. Did you already have once a venereal disease, sexually transmitted, hepatitis/jaundice? Were you once suspected of being infected with HIV	4
Risky behaviors	14
8. Do you belong to a risk group for HIV infection (injection of drugs, sexual contacts without condom, traditional vaccines with blades used for different persons, several blood transfusions)?	14
Medical history	54
9. Do you suffer from diseases of the cardiovascular system (e. g pericardial pain)?	0
10. Do you suffer from diseases of the respiratory system (e.g., asthma)?	0
11. Do you suffer from diseases of the gastrointestinal system (e.g.; peptic ulcer)	0
12. Do you suffer from diseases of the central nervous system (e.g. epilepsy)?	0
13. Do you suffer from diseases of the genital/urinary system?	0
14. Do you suffer from diseases of the endocrine system? (e.g. diabetes).	0
15. Do you suffer from diseases of the immune system? (e.g. allergy)	0
16. Do you have a blood disease? (e.g. anemia, hemophilia)?	7
17. Do you have a fever?	9
18. Are you pregnant or menstruating? (Women)	29
19. Are you taking any medication?	9
20. Were you vaccinated in the last 4 weeks?	0
Physical examination	Total number of persons excluded: 79
21. Blood pressure higher than 160/110 mmHg or lower than 120/80 mmHg?	59
22. Other (Under and over weight; under age)	20

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4. Discussion

Blood transfusion is a life-saving procedure, which carries the risk of transmitting life threatening infectious such as HIV, hepatitis B, and hepatitis C, which are major public health problems in developing countries. Blood transfusion being a potential transmission vehicle of these infections, identifying people at risk and knowing the prevalence of transfusion-transmitted infectious diseases (TTIs) among blood donors is crucial in choosing blood transfusion deferral policies and infection control strategies [12,13].

The present study revealed a seroprevalence among the donors of 2.5%, 2.5%, 1.1%, and 3.3% for HIV, HBV, HCV and TPHA markers respectively. Compared to other African countries the overall prevalence of TTIs among blood donors is relatively low [10,14].

Accuracy and honesty are essential for the blood donors' questionnaire to increase transfusion safety. In Africa people are not always comfortable giving private information about themselves. Therefore, the reliability and effectiveness of medical questionnaire in the selection of blood donors is a problem that needs to be assessed. The present report is one of the few evaluating in sub-Saharan Africa the benefit of blood donors' questionnaire.

In the Gabonese setting, medical status and risky behaviors were the principal reasons for blood donation deferral. Data showed that the blood donors' screening questionnaire excluded 18.7% of potential donors from blood donation and reduced by 0.6%, 0.35% and 1.3% respectively the proportion of HBV, HCV and TPHA seroreactive donors being selected for donation. The questionnaire deferral rate in Gabon was higher than what was observed in Mozambique (18.7% vs 9.5%) [10]. However like in Mozambique our study showed that the blood donors' screening questionnaire does not exclude effectively HIV-infected donors. One of the possible explanations for that is that in the African context were the stigma around HIV and sexual practices or behaviors still hangs; people are not always forward when it comes to disclose their HIV status or risky behaviors [15,16]. We should think of a way to ameliorate the comfort level of African blood donors with questions addressing sexual risk behaviors.

Despite the lake of effectiveness regarding HIV screening of donors, the blood donor questionnaire remains important in improving safe blood collection. Another possible advantage of the blood donor-screening questionnaire would be the reduction of undetermined diagnostic test results. This is important, as the questionnaire would reduce the waste of expensive and valuable kits, which is not negligible in a developing world setting. Although blood donor questionnaire reduced the risk for TTIs, surveillance of blood transfusion recipients and epidemiological TTIs data should be available for the country to design appropriate deferral policies directed at these events.

5. Conclusion

The prevalence of HBV, HCV and syphilis, were reduced in donors receiving a donor history questionnaire, but there was no reduction in HIV marker prevalence.

Disclosure of interest

The authors declare that they have no conflicts of interest concerning this article.

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